# ZHANHONG HE



PhD, Dec 2026 Graduation, University of Western Australia (UWA Profile)

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## Research Profile

My PhD research focuses on expressive musical dynamics, audio-symbolic modelling and music generation, including MIDI velocity estimation, score-informed transcription refinement, multitask learning and human perceptual evaluation. My industrial research experience at Dolby and Huawei spans music generation, reinforcement learning and spatial audio. This is complemented by university teaching and project co-supervision in electronic engineering and computer science. My music coursework and practical music-making experience also motivate my interest in developing tools that give musicians intuitive control over musical expression.

## Education

### University of Western Australia

### PhD research in Electrical and Electronic Engineering | February 2022-December 2026

Doctoral defense completed; graduation certificate expected November 2026. Formal award date pending confirmation.

Research focus: musical dynamics, audio and symbolic music processing, and machine learning.

University International Fee Scholarship; University Postgraduate Awards; 2026 UWA Data Institute Award.

### University of Western Australia

### BSc (Honours) in Engineering Science | October 2018-December 2021

First Class Honours; GPA 7/7; WAM 87.7/100.

Honours thesis: Improving Spectrogram-based Music Source Separation using Phase Information.

## Selected Research

Expressive dynamics and audio-symbolic modelling (University of Western Australia)

- Developed score-informed Transformer models for MIDI velocity refinement and multitask CRNNs for jointly estimating musical dynamics, beats, downbeats and structural changes.
- Adapted piano-pretrained velocity estimation to other instruments using differentiable SoundFont proxies and loudness-based supervision, reducing reliance on note-level velocity labels.

LLM post-training for music generation (Huawei Hong Kong)

- Applied DPO and GRPO to LLM-based music generation, including vocal-to-accompaniment and lyrics-to-music tasks
- Developed music-specific reward signals and rollout workflows, and collected and analysed human-preference data for model evaluation and post-training.

Choral transcription and agentic data workflows (Great Bay University)

- Conducted research on choral singing transcription, including note transcription and vocal-part assignment.
- Built agentic workflows for data collection and dataset construction, incorporating ACE Studio synthesis.

Generative modelling and perceptual evaluation (Dolby Australia)

- Explored U-Net, VAE and masked Transformer architectures for music generation, including a U-Net approach to per-note MIDI velocity prediction. Designed a listening test with professional listeners.

## Research Experience

### Huawei Hongkong Research Centre - Research Intern, full time | Jun-Dec 2026

- Conducted research on music-model post-training and human-preference evaluation.

### Dolby Laboratories, Sydney - Research Intern, full time | May-September 2024

- Developed music-generation models using U-Net, VAE and masked Transformer architectures.
- Designed and conducted professional listening tests to evaluate generated musical expression.

### Great Bay University - Visiting Student Researcher, part time | Feb-Dec 2026

- Explored pretrained music models (e.g.,MERT) and agentic workflows for music research.

## Teaching And Collaboration

### University of Western Australia - Tutor and Lab Facilitator, part time | Feb 2023-Dec 2025

- Supported ELEC4404 Signal Processing and GENG2000 Engineering Practices.
- Co-supervised CITS4011 Computer Science Honours projects and final-year designs.

### Foreign Trade University, HCMC, Vietnam – Teaching Assistant, part time | Jun 2019-Jul 2019

- Held 2-month Language and Culture Summer Program for local students.

## Publications And Preprints

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 **Zhanhong He**, Hanyu Meng and Yaolong Ju. CASM: Context-Aware Semi-Markov Post-Processor for Beat Tracking. Under review in ICASSP2027. [Demo] [Arxiv]

 Hanyu Meng, **Zhanhong He**, Zixun Guo and Yaolong Ju. Toward Part-Aware Choral Transcription with singing voice assignment. Under review in ICASSP2027. [Demo] [Arixv]

 Hanyu Meng, Hao Shi, **Zhanhong He**, et al. Speaker-Consistent Grounding for Confusion-Resilient LLM-Based Target Speech Extraction. Under review in ICASSP2027.  Mingda Song, Yueqiao Zhang, Yulun

 Yulun Wu, **Zhanhong He**, et al. HCMAMBA: Harmonically Informed State-Space Modeling for Large-Vocabulary Audio Chord Recognition. Under review in ICASSP2027.

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 **Zhanhong He**, Hanyu Meng, Jiahao Zhao, Ganghui Ru and Yaolong Ju. MIREX 2026 Submission: Context-Aware Semi-Markov Post-Processor for Beat and Downbeat Estimation. ISMIR Late breaking Demo, 2026. [MIREX]

 Hanyu Meng, **Zhanhong He** and Yaolong Ju. PawChorale: A Scalable Synthetic Dataset for Part-Aware Choral Transcription and Separation. ISMIR Late breaking Demo, 2026. [Dataset]

 **Zhanhong He,** Hanyu Meng, David Defeng Huang and Roberto Togneri. Beyond Piano: Cross-Instrument MIDI Velocity Estimation via Differentiable SoundFont Proxies. Int. Soc. Music Info. Retrieval (ISMIR), 2026. [GitHub]

 **Zhanhong He**, Hanyu Meng, Defeng (David) Huang and Roberto Togneri. Joint Estimation of Piano Dynamics and Metrical Structure with a Multi-Task Multi-Scale Network. IEEE Int. Conf. Acoust., Speech, Signal Process. (ICASSP), 2026. [GitHub]

 **Zhanhong He,** David Cooper, Defeng Huang and Roberto Togneri. Filling MIDI Velocity using U-Net Image Colorizer. Int. Symposium on Computer Music Multidisciplinary Research (CMMR), 2025. [GitHub]

## Music Practice and Interactive Systems

- **UWA Conservatorium of Music** coursework in electronic music, music generation and ensemble performance, 2023; composition and mixing music with Ableton Live [Demo Video], and participation in a public ensemble concert [YouTube].
- Built a coursework HMM system for automatic piano and environmental-sound generation system, using Max/MSP with Ableton Live.
- Built an Arduino & PCB-based theremin with a MIDI interface, programmed in C, at a Physical Interaction Design for Music workshop, Stanford University CCRMA, 2023.

## Technical Skills

Python and PyTorch; LLM Post-Training; DDSP, U-Net and VAE research; audio and symbolic music processing; human perceptual evaluation; Ableton Live and Max/MSP.

Languages: English (fluent); Chinese and Cantonese (native).